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GEN-06—Application of new specifications and approval of trade measurement devices that incorporate technologies not covered by the existing specifications

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1.0 Purpose

The purpose of this bulletin is to establish the policy for the enforcement of new specifications developed under the authority of the *Weights and Measures Act* (WMA) or the *Electricity and Gas Inspection Act* (EGIA), as well as the policy for the approval and inspection of trade measurement devices that incorporate technologies not covered by the existing specifications enforced by Measurement Canada (MC).

2.0 Scope

This bulletin applies to specifications and requirements developed to address devices and components under the WMA and meters under the EGIA. For reasons of simplicity, they are referred to in this document as trade measurement devices.

3.0 References

- Electricity and Gas Inspection Act (R.S. 1985, c. E-4), subsection 9(4).
- Electricity and Gas Inspection Regulations (SOR/86-131), section 15.
- GEN-04—Notices of conditional approval—naming convention, terms and conditions, disclosure, notification, initial inspection, and inspection certificates
- GEN-26—Modifications to approved weighing and measuring devices and electricity and gas meters
- GEN-38—Principles and prerequisites for approval of electricity and gas metering devices incorporating new technology
- Weights and Measures Act (R.S.C., 1985, c. W-6), subsection 3(2).

4.0 Effective date of specifications

4.1 Specifications developed under the authority of the WMA are effective after they have been published in the *Canada Gazette*. Unless otherwise specified, the effective date of a new specification will coincide with the publication date in the *Canada Gazette*.

4.2 Unless otherwise specified, specifications developed under the authority of the EGIA are effective after they have been authorized by the President of MC.

4.3 Once specifications become effective, accredited organizations shall, where applicable, revise their quality assurance documentation to meet the requirements of the new specifications.

4.4 Notwithstanding sections 4.1 and 4.2, trade measurement devices incorporating technologies not covered by the existing specifications may be conditionally approved and inspected against the terms and conditions or provisional specifications, as may be applicable, defined in section 7.0 of this document and listed in Appendix A.

5.0 Application of new specifications for the purpose of approval and inspection

5.1 Trade measurement devices approved on or after the effective date of new specifications

Unless otherwise specified, trade measuring devices approved on or after the effective date of the new specifications will be evaluated against those specifications in their entirety. Trade measuring devices submitted for approval before the effective date of the specifications, but evaluated and/or approved after that date, will also be evaluated against the new specifications in their entirety.

5.2 Trade measurement devices approved before the effective date of new specifications but initially inspected or verified after that date

Unless otherwise specified, trade measurement devices approved before the effective date of the specifications, but initially inspected or verified after that date, are not required to meet the technical requirements (e.g., design, composition and construction) or the administrative requirements (e.g., marking) of the new specifications. In those cases, the devices will

continue to be evaluated against the technical and administrative requirements that were in force when the devices were approved. However, the metrological performance (e.g., limits of error/tolerance) and the installation and use requirements prescribed by the new specifications are applicable.

5.3 Trade measurement devices approved and initially inspected or verified prior to the effective date of new specifications

5.3.1 Unless otherwise specified, trade measurement devices that have been approved and initially inspected or verified prior to the effective date of the specifications are not required to meet the technical requirements (e.g., design, composition and construction) or the administrative requirements (e.g., marking) of the new specifications.

5.3.2 For devices covered under the WMA, the metrological performance (e.g., limits of error/tolerance) and use requirements prescribed by the new specifications are applicable unless otherwise specified. Installation requirements need not be met until the affected device is removed and reinstalled elsewhere or a new device is installed in its place.

5.3.3 For devices covered under the EGIA, the metrological performance (e.g., limits of error/tolerance) and the installation and use requirements prescribed by the new specifications are applicable unless otherwise specified.

5.4 Trade measurement devices modified and resubmitted to the Approval and Calibration Services Laboratory for evaluation after the effective date of new specifications

Trade measurement devices that are modified in a manner that requires them to be resubmitted to the Approval and Calibration Services Laboratory for a complete or partial evaluation (see GEN-26) will be

evaluated against the metrological, technical and administrative requirements prescribed by the specifications before a new or revised Notice of Approval (NOA) can be issued.

6.0 Special considerations

There may be circumstances where the requirements of a new specification will have to be applied differently than the manner described in section 5.0 of this document. In these instances, the specification will include a section which prescribes the manner in which the requirements are to be applied.

7.0 Approval and inspection of trade measurement devices that incorporate technologies not covered by the existing specifications

7.1 Application of terms and conditions for the purposes of approval and inspection pursuant to the *Weights and Measures Act*

7.1.1 In order to approve trade measurement devices that incorporate technologies not covered by the existing specifications without incurring lengthy delays which can be associated with the regulatory process, trade measurement devices subject to the WMA may be evaluated against terms and conditions that have been authorized by the President of MC. Table 1 of Appendix A contains the list of the related terms and conditions.

7.1.2 When a new type of trade measurement device is submitted for approval, the Program Development Directorate (PDD) may, in consultation with the Engineering and Laboratory Services Directorate (ELSD), the approval applicant and other affected parties, develop terms and

conditions applicable to the subject device in order to allow for its approval and inspection. New trade measurement devices shall be examined and tested for compliance against these requirements.

7.1.3 Devices that comply with the authorized terms and conditions listed in table 1 of Appendix A shall be temporarily approved pursuant to subsection 3(2) of the WMA. The notices of temporary approval are known as notices of conditional approval (NOCAs).

7.1.4 NOCAs will remain in effect until the applicable specification has been formally adopted through the regulatory process.

7.1.5 NOCAs will indicate, directly or by reference, the terms and conditions to which the new device type must conform.

7.1.6 There may be instances where trade measurement devices that have been initially inspected or verified under a temporary or conditional approval will require modification to meet the applicable regulations and specifications once formally adopted. This information will be incorporated into the device's NOCA.

7.1.7 PDD will inform all interested parties of the application of new requirements or exemptions for a category or type of device.

7.1.8 Device inspections covered by a NOCA granted in accordance with the terms and conditions specified pursuant to subsection 3(2) of the WMA will be conducted on the basis of the requirements set out in said terms and conditions. The data resulting from these inspections will be used to evaluate devices pending final approval as described in GEN-04.

7.2 Application of provisional specifications for the purposes of approval and inspection pursuant to the *Electricity and Gas Inspection Act*

7.2.1 In order to approve trade measurement devices that incorporate technologies not covered by the existing specifications without incurring lengthy delays which can be associated with public consultation at the national level, trade measurement devices subject to the EGIA may be evaluated against provisional specifications that have been authorized by the President of MC. Table 2 in Appendix A contains the list of specific provisional specifications under which devices can only be granted a NOCA.

7.2.2 When a new type of trade measurement device is submitted for approval, PDD may, in consultation with the ELSD, the approval applicant and other affected parties, develop provisional specifications applicable to the subject device in order to allow for its approval and inspection. New trade measurement devices shall be examined and tested for compliance against these requirements.

7.2.3 Devices that comply with the requirements of the provisional specifications listed in table 2 in Appendix A shall be issued a NOCA pursuant to subsection 9(4) of the EGIA and section 15 of the *Electricity and Gas Inspection Regulations*.

7.2.4 NOCAs will remain in effect until the applicable provisional specification has been reviewed and revised by MC, undergone public consultation with industry participation, finalized and authorized by MC as a specification.

7.2.5 NOCAs will indicate, directly or by reference, the provisional specifications to which the new device type must conform.

7.2.6 There may be instances where trade measurement devices that have been initially inspected or verified under a temporary or conditional approval will require modification to meet the applicable regulations and

final specification once they have undergone public/industry consultation and been authorized by MC. This information will be included as part of the trade measurement device's NOCA.

7.2.7 PDD will inform all interested parties of the application of new requirements or exemptions for a category or type of device.

8.0 Revisions

The effective date of the initial version of this bulletin was 1993-09-01.

8.1 The purpose of revision 16 was to remove references to the following provisional specifications:

- PS-G-15—Provisional specifications for the approval, verification, installation and use of fluidic oscillation meters
- PS-G-16—Provisional specifications for the approval, verification, reverification, installation and use of conditioning orifice plates

8.2 The purpose of revision 15 (2016-05-19) was to:

a. Make minor editorial and format changes to clearly identify provisional specifications under which devices can only be granted a NOCA.
b. Create a Reference section which adds bulletins GEN-04, GEN-26 and GEN-38.
c. Remove the following provisional specifications:
    - PS-E-14 — Provisional specifications for the verification, reverification, installation and use of NXCT®electronic current transformer
    - PS-E-16 — Provisional specifications for the verification, reverification, installation and use of NXVT®electronic voltage transformer

- PS-G-13 — Provisional specifications and procedures for the approval of correction devices and linearization functions incorporated in meters and flow computers
- PS-G-14 — Provisional specifications and procedures for the verification of correction devices and linearization functions incorporated in meters and flow computers

d. Divide Appendix A into two tables:

- Table 1 — Authorized terms and conditions (*Weights and Measures Act*)
- Table 2 — Authorized provisional specifications (*Electricity and Gas Inspection Act)*

8.3 The purpose of revision 14 (2015-03-06) was to correct section 4.3 regarding previously inspected devices to indicate that installation requirements for W&M devices need not be met when new requirements are imposed unless the device is subsequently removed from the initially inspected installation and reinstalled elsewhere. This is consistent with previous versions (prior to version 10) of the bulletin. E&G devices may need to be brought into compliance immediately upon adoption of the new requirements.

Appendix A

The following tables list the terms and conditions and provisional specifications under which devices can only be granted Notices of Conditional Approval.

Table 1 – Authorized terms and conditions

Terms and conditions	Mass and dimension	Volume and time
Terms and conditions for the approval of timber dimension measuring devices	✔	-
Terms and conditions for the approval of multiple dimension measuring devices	✔	-
Terms and conditions for the approval of metrological audit trails	✔	✔
Terms and conditions for the approval of Coriolis liquid meters	-	✔
Terms and conditions for the approval of pressure transducers	-	✔
Terms and conditions for the approval of electronic APC incorporated into electronic registers	-	✔

Table 2 – Authorized provisional specifications

Provisional specifications	Electricity	Gas
PS-E-13—Provisional specifications for the approval of electronic current transformers	✔	-
PS-E-15—Provisional specifications for approval of electronic voltage transformers	✔	-
PS-EG-01—Provisional specifications for the approval of prepayment meters	✔	✔

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